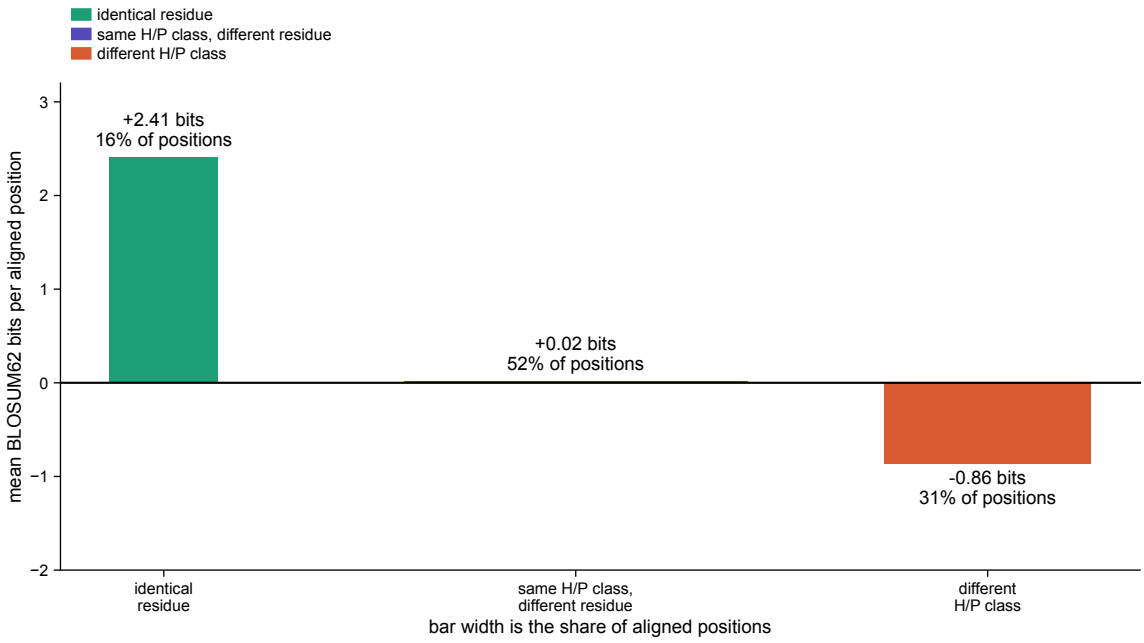
TOOLS: no kmerseek; Pfam-A 38.2 seed pairs from notebook 230 at 20% identity or less; BLOSUM62 (Biopython), hp_thomas_dill2 classes

Half of the aligned positions in distant Pfam pairs keep their H/P class with a different residue, and BLOSUM62 scores them at zero



Hypothesis: If BLOSUM62 already credits a kept H/P class, the middle bar is well above zero.

Conclusion: identical residue: 16% of positions, +2.41 bits. same H/P class, different residue: 52% of positions, +0.02 bits. different H/P class: 31% of positions, -0.86 bits.